

Scott Workman

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ORCID: 0000-0002-7145-7484
Publications: 2,100+ citations, h-index 19
Citizenship: United States

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Areas of Expertise

Computer Vision, Deep Learning, Generative AI, 3D/Geometric Vision, Multimodal Fusion, Geospatial AI, Self-Supervised Learning

Education

University of Kentucky Ph.D. in Computer Science Adviser: Nathan Jacobs Thesis: Leveraging Overhead Imagery for Localization, Mapping, and Understanding	2012–2018
University of Kentucky B.S. in Computer Science <i>Magna Cum Laude</i> with Honors	2006–2010

Appointments

Research Scientist & Consultant Independent	2025–Present <i>Remote</i>
• Providing strategic consulting on geospatial AI and computer vision for government and industry partners. • Conducting research on topics such as neural planetary representations and geospatial world modeling, with publications including mixed-view panorama synthesis [30] and manuscripts in preparation. • Collaborating with researchers at Washington University in St. Louis and Temple University on cross-view scene understanding, image-driven mapping, and multimodal fusion. • Maintaining and extending open-source datasets and benchmarks widely adopted by the research community.	
Principal Research Scientist DZYNE Technologies	2021–2025 <i>Fairfax, VA</i>
• Led and managed a multidisciplinary research team, overseeing the full R&D pipeline from data processing and distributed model training on HPC clusters to deployment, with a focus on machine learning and computer vision, while maintaining an active, hands-on role. • Drove research initiatives, developed novel methodologies, and produced findings that contributed to publications, presentations, and grant proposals. • Managed project development, actively engaging in experimental design, testing, and presenting results to both technical and non-technical stakeholders. • Mentored junior researchers, supporting their growth and development.	

- Contributed to securing over \$5M in funding for research initiatives, including serving as Co-PI/Co-I on projects funded by agencies such as DARPA, IARPA, and NGA.

Senior Research Scientist

DZYNE Technologies

2019–2021

Fairfax, VA

- Conducted independent research, developed novel algorithms, and contributed to technical strategy and problem-solving across multiple projects, collaborating with cross-disciplinary teams.
- Led and co-authored publications on topics such as pose estimation [5, 34], depth estimation [6], multi-modal fusion [2, 4], and image-driven traffic modeling [1, 9], in collaboration with academic and institutional partners.

Research Assistant

Multimodal Vision Research Laboratory, University of Kentucky

2011–2018

Lexington, KY

- Developed multimodal fusion techniques for joint inference across various modalities, including ground-level imagery, overhead imagery, and audio, to improve geospatial world models [12, 14, 15].
- Introduced the first method for cross-view image synthesis, using weak supervision from labeled ground-level imagery to enhance overhead image understanding [16].
- Developed learning-based methods for directly estimating geometric properties of images, such as the horizon line and focal length [17, 24].
- Leveraged overhead imagery to support ground-level image localization, introducing the CVUSA dataset [23].
- Estimated 3D scene geometry for a static outdoor camera, or network of cameras, using videos from several partly cloudy days [28, 33].
- Demonstrated how a single image of a rainbow can be used for camera calibration and image geolocalization [26].
- Estimated radial distortion, focal length, and geo-orientation using long-term observations of outdoor scenes [29].

Research Assistant

Center for Educator Preparation Information Systems, University of Kentucky

2010–2011

Lexington, KY

- Engaged in research and development on information systems applied to educator preparation.
- Developed an advising management system used by students and faculty of the College of Education.

Software Engineer Intern

CorrectCare - Integrated Health

Summer 2010

Lexington, KY

Network Engineer Intern

Network Engineering Management and Operations Center, University of Kentucky

2008–2010

Lexington, KY

IT Support Specialist

Engineering Computing Services, University of Kentucky

2007–2010

Lexington, KY

Family Business

Deer Run Stables & Horseman's Retreat

1997–Present

Richmond, KY

Honors and Awards

- **Research Contributions**
 - *Leadership and Service*

- * Lead Area Chair: CVPR 2026
- * Area Chair: CVPR, ICCV, ECCV, NeurIPS, ICML, WACV 2024–2026
- *Service Recognitions*
 - * Outstanding Area Chair Recognition:
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
 - * Outstanding Reviewer Recognition:
 - Conference on Neural Information Processing Systems (NeurIPS) 2023
 - European Conference on Computer Vision (ECCV) 2020
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2020–2022
 - IEEE/CVF International Conference on Computer Vision (ICCV) 2019
- *Selected Presentations*
 - * Oral Presentation, CVPR (top 5% of submissions) 2020
- **Prestigious Programs**
 - Young Researcher, 5th Heidelberg Laureate Forum, Heidelberg, Germany 2017
 - Young Researcher, International Computer Vision Summer School, Punta Sampieri, Sicily 2014
- **Academic Excellence**
 - Outstanding Ph.D. Student in Computer Science, University of Kentucky 2017
 - Dean’s List, University of Kentucky 2008–2010
- **Fellowships & Grants**
 - Burton E. Heard Graduate Fellowship 2016–2017
 - NVIDIA Academic Hardware Grant (Tesla K40) 2015
 - Presidential Fellowship (nominated), University of Kentucky 2015
 - Alltel/Windstream Scholarship, University of Kentucky 2008–2010

Publications

Summary: 45+ publications including 29 peer-reviewed conference papers, 4 journal articles, 6 workshop papers, and 1 US patent at venues such as CVPR, ICCV, and ECCV.

Conference Papers (Refereed)

- [1] Scott Workman and Armin Hadzic. Probabilistic Image-Driven Traffic Modeling via Remote Sensing. In *European Conference on Computer Vision (ECCV)*, 2024.
- [2] Connor Greenwell, Jon Crall, Matthew Purri, Kristin Dana, Nathan Jacobs, Armin Hadzic, Scott Workman, and Matt Leotta. WATCH: Wide-Area Terrestrial Change Hypercube. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2024.
- [3] Scott Workman, Armin Hadzic, and M. Usman Rafique. Handling Image and Label Resolution Mismatch in Remote Sensing. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2023.
- [4] Scott Workman, M. Usman Rafique, Hunter Blanton, and Nathan Jacobs. Revisiting Near/Remote Sensing with Geospatial Attention. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [5] Hunter Blanton, Scott Workman, and Nathan Jacobs. A Structure-Aware Method for Direct Pose Estimation. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [6] Scott Workman and Hunter Blanton. Augmenting Depth Estimation with Geospatial Context. In *IEEE International Conference on Computer Vision (ICCV)*, 2021.
- [7] Scott Workman, M. Usman Rafique, Hunter Blanton, Connor Greenwell, and Nathan Jacobs. Single Image Cloud Detection via Multi-Image Fusion. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2020. Oral.

- [8] Tawfiq Salem, Scott Workman, and Nathan Jacobs. Learning a Dynamic Map of Visual Appearance. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [9] Scott Workman and Nathan Jacobs. Dynamic Traffic Modeling from Overhead Imagery. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. Oral.
- [10] Menghua Zhai, Tawfiq Salem, Connor Greenwell, Scott Workman, Robert Pless, and Nathan Jacobs. Learning Geo-Temporal Image Features. In *British Machine Vision Conference (BMVC)*, 2018.
- [11] Connor Greenwell, Scott Workman, and Nathan Jacobs. What Goes Where: Predicting Object Distributions from Above. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2018.
- [12] Tawfiq Salem, Menghua Zhai, Scott Workman, and Nathan Jacobs. A Multimodal Approach to Mapping Soundscapes. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2018.
- [13] Weilian Song, Scott Workman, Armin Hadzic, Xu Zhang, Eric Green, Mei Chen, Reginald Souleyrette, and Nathan Jacobs. FARSA: Fully Automated Roadway Safety Assessment. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018.
- [14] Scott Workman, Menghua Zhai, David J. Crandall, and Nathan Jacobs. A Unified Model for Near and Remote Sensing. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [15] Scott Workman, Richard Souvenir, and Nathan Jacobs. Understanding and Mapping Natural Beauty. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [16] Menghua Zhai, Zach Bessinger, Scott Workman, and Nathan Jacobs. Predicting Ground-Level Scene Layout from Aerial Imagery. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [17] Scott Workman, Menghua Zhai, and Nathan Jacobs. Horizon Lines in the Wild. In *British Machine Vision Conference (BMVC)*, 2016.
- [18] Menghua Zhai, Scott Workman, and Nathan Jacobs. Camera Geo-Calibration using an MCMC Approach. In *International Conference on Image Processing (ICIP)*, 2016.
- [19] Menghua Zhai, Scott Workman, and Nathan Jacobs. Detecting Vanishing Points using Global Image Context in a Non-Manhattan World. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. Spotlight.
- [20] R. Paul Mihail, Scott Workman, Zach Bessinger, and Nathan Jacobs. Sky Segmentation in the Wild: An Empirical Study. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2016.
- [21] Ryan Baltenberger, Menghua Zhai, Connor Greenwell, Scott Workman, and Nathan Jacobs. A Fast Method for Estimating Transient Scene Attributes. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2016.
- [22] Tawfiq Salem, Scott Workman, Menghua Zhai, and Nathan Jacobs. Analyzing Human Appearance as a Cue for Dating Images. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2016.
- [23] Scott Workman, Richard Souvenir, and Nathan Jacobs. Wide-Area Image Geolocalization with Aerial Reference Imagery. In *IEEE International Conference on Computer Vision (ICCV)*, 2015.
- [24] Scott Workman, Connor Greenwell, Menghua Zhai, Ryan Baltenberger, and Nathan Jacobs. DeepFocal: A Method for Direct Focal Length Estimation. In *International Conference on Image Processing (ICIP)*, 2015.
- [25] Mohammad T. Islam, Scott Workman, and Nathan Jacobs. Face2GPS: Estimating Geographic Location from Facial Features. In *International Conference on Image Processing (ICIP)*, 2015. Oral.

- [26] Scott Workman, R. Paul Mihail, and Nathan Jacobs. A Pot of Gold: Rainbows as a Calibration Cue. In *European Conference on Computer Vision (ECCV)*, 2014.
- [27] Mohammad T. Islam, Scott Workman, Hui Wu, Richard Souvenir, and Nathan Jacobs. Exploring the Geo-Dependence of Human Face Appearance. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2014.
- [28] Nathan Jacobs, Scott Workman, and Richard Souvenir. Scene Geometry from Several Partly Cloudy Days. In *ACM/IEEE International Conference on Distributed Smart Cameras (ICDSC)*, 2013.
- [29] Nathan Jacobs, Mohammad T. Islam, and Scott Workman. Cloud Motion as a Calibration Cue. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2013.

Journal Articles (Refereed)

- [30] Zhexiong Xiong, Xin Xing, Scott Workman, Subash Khanal, and Nathan Jacobs. Mixed-View Panorama Synthesis using Geospatially Guided Diffusion. *Transactions on Machine Learning Research (TMLR)*, 2025.
- [31] Rafael Padilha, Tawfiq Salem, Scott Workman, Fernanda A. Andaló, Anderson Rocha, and Nathan Jacobs. Content-Aware Detection of Temporal Metadata Manipulation. *IEEE Transactions on Information Forensics & Security (TIFS)*, 17:1316–1327, March 2022.
- [32] Nathan Jacobs, Scott Workman, and Richard Souvenir. Cloudmaps from Static Ground-View Video. *Image and Vision Computing (IVC)*, 52:154–166, August 2016.
- [33] Scott Workman, Richard Souvenir, and Nathan Jacobs. Scene Shape Estimation from Multiple Partly Cloudy Days. *Computer Vision and Image Understanding (CVIU)*, 134:116–129, May 2015.

Workshop Papers (Refereed)

- [34] Hunter Blanton, Connor Greenwell, Scott Workman, and Nathan Jacobs. Extending Absolute Pose Regression to Multiple Scenes. In *CVPR Workshop: Long-Term Visual Localization, Visual Odometry and Geometric and Learning-based SLAM*, 2020.
- [35] Armin Hadzic, Hunter Blanton, Weilian Song, Mei Chen, Scott Workman, and Nathan Jacobs. RasterNet: Modeling Free-Flow Speed using LiDAR and Overhead Imagery. In *CVPR Workshop: Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION)*, 2020.
- [36] Connor Greenwell, Scott Workman, and Nathan Jacobs. Implicit Land Use Mapping Using Social Media Imagery. In *IEEE Applied Imagery Pattern Recognition Workshop (AIPR)*, 2019.
- [37] Tawfiq Salem, Menghua Zhai, Scott Workman, and Nathan Jacobs. A Multimodal Approach to Mapping Soundscapes. In *CVPR Workshop: Sight and Sound*, 2018.
- [38] Nathan Jacobs, Scott Workman, and Menghua Zhai. Cross-view Convolutional Networks. In *IEEE Applied Imagery Pattern Recognition Workshop (AIPR)*, 2016.
- [39] Scott Workman and Nathan Jacobs. On the Location Dependence of Convolutional Neural Network Features. In *IEEE/ISPRS Workshop: Looking from above: When Earth observation meets vision (EARTHVISION)*, 2015.

Abstracts

- [40] Scott Workman and Nathan Jacobs. Scene Understanding using Clouds. In *International Computer Vision Summer School (ICVSS)*, 2014.
- [41] J. David Smith, Ryan Baltenberger, Scott Workman, and Nathan Jacobs. User-in-the-Loop Calibration and Mensuration. In *National Conference on Undergraduate Research (NCUR)*, 2014.

- [42] Ryan Baltenberger, James Knochelmann, Scott Workman, Mohammad T. Islam, Nathan Jacobs, and James Griffioen. Constructing a High-Resolution Mosaic of Kentucky Lake. In *Kentucky GIS Conference*, 2013. Best Student Presentation.
- [43] XuZi Zhou, Scott Workman, Mohammad T. Islam, Nathan Jacobs, and James Griffioen. Cyber Infrastructure for the VOEIS Project. In *Symposium in the Mathematical, Statistical and Computer Sciences*, 2013. Best Student Presentation.
- [44] Scott Workman, James Knochelmann, Nathan Jacobs, David S. White, and Richard Hauer. Registration and Visualization of Scientific Aerial Imagery at Kentucky Lake. In *Kentucky EPSCoR Conference*, 2012.

Patents

- [45] Nathan Jacobs and Scott Workman. Network architecture for generating a labeled overhead image. US Patent 10,755,146, August 2020.

Datasets & Benchmarks

CVUSA (Cross-View USA)	1.58M matched ground-to-aerial image pairs for cross-view geolocalization, image synthesis, and navigation.	[23]
HLW (Horizon Lines in the Wild)	100K+ natural images annotated with horizon line positions via Structure-from-Motion.	[17]
Brooklyn & Queens	53K+ overhead tiles paired with 175K+ street-level panoramas for joint near/remote sensing across five urban labeling tasks.	[4, 14]
Dynamic Traffic Speeds	Overhead imagery linked with historical traffic speed records for NYC and Cincinnati, enabling image-driven traffic modeling.	[1, 9]
CVT (Cross-View Time)	Ground-level and overhead imagery linked with temporal meta-data for learning dynamic maps of visual appearance.	[8]
HoliCity-Overhead	Ground-level images paired with overhead imagery and digital surface models for augmenting monocular depth estimation.	[6]
Low2High	Benchmark for resolution mismatch in remote sensing, pairing 1m imagery with coarse 30m labels and high-resolution exemplars.	[3]
CVSoN (Cross-View ScenicOrNot)	Dataset for mapping perceived natural beauty from ground-level and overhead imagery.	[15]
SkyFinder	Webcam images annotated with pixel-level sky masks for sky segmentation in the wild.	[20]
Face2Year	Images extracted from high-school yearbooks for analyzing temporal patterns in human appearance.	[22]
San Francisco	Geo-tagged image dataset for studying the location dependence of CNN features.	[39]

Professional Service

- Lead Area Chair:
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2026]
- Area Chair:

- European Conference on Computer Vision (ECCV) [2026]
- IEEE/CVF International Conference on Computer Vision (ICCV) [2025]
- International Conference on Machine Learning (ICML) [2025, 2026]
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2025, 2026]
- Conference on Neural Information Processing Systems (NeurIPS) [2024-2026]
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2024, 2025]
- Reviewing for Journals:
 - ISPRS Journal of Photogrammetry and Remote Sensing (P&RS) [2024]
 - Computer Vision and Image Understanding (CVIU) [2019]
 - IEEE Transactions on Multimedia (TMM) [2016]
 - EURASIP Journal on Image and Video Processing (JIVP) [2015]
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS) [2015]
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI) [2015, 2016]
- Program Committee / Reviewer for:
 - Conferences
 - * International Conference on Machine Learning (ICML) [2024]
 - * Conference on Neural Information Processing Systems (NeurIPS) [2022, 2023]
 - * International Conference on Learning Representations (ICLR) [2022-2024]
 - * IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2022-2024]
 - * British Machine Vision Conference (BMVC) [2019, 2020]
 - * ACM SIGGRAPH [2018]
 - * European Conference on Computer Vision (ECCV) [2014, 2020, 2022, 2024]
 - * IEEE/CVF International Conference on Computer Vision (ICCV) [2013, 2019, 2021, 2023]
 - * IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2013-2023]
 - Workshops
 - * IEEE/CVF Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019-2026]
 - * ACM Workshop on Geotagging and Its Applications in Multimedia (GeoMM) [2013, 2014]
- Volunteer / Presenter:
 - Graduate Student Representative, Association for Computing Machinery, UK Student Chapter [2014-2015]
 - Engineers Day (E-Day), University of Kentucky [2013-2014]

Grants & Funded Projects

Leadership Roles

- | | |
|--|-------------|
| • NGA Boosting Innovative GEOINT Research (BIG-R) | \$3,000,000 |
| Proposal: <i>Terrain Mobility Intelligence (TMI)</i> | |
| Co-PI, DZYNE Technologies | 2022–2025 |

- **IARPA Space-based Machine Automated Recognition Technique (SMART)** \$2,000,000
Proposal: *WATCH: Wide-Area Terrestrial Change Hypercube* (Phase 1–3)
Co-PI, DZYNE Technologies (Prime: Kitware) 2021–2024
- **Army NVESD Long-Range Depth Estimation for Autonomous Vehicles** \$850,000+
Proposal: *GIS-Enabled Vehicle Video Depth Mapping*
Co-PI, DZYNE Technologies 2020–2021
- **DARPA SBIR - Software/Analytics Exploiting Commercial Satellite Imagery** \$1,500,000+
Proposal: *Cloud-Based Analytics Of Satellite Imagery* (Phase 2–2.5)
Co-I, DZYNE Technologies 2020–2022
- **DARPA Geospatial Cloud Analytics (GCA)** \$2,000,000
Proposal: *Cloud Analytics of Geospatial Imagery (CAGI)* (Phase 2–2.5)
Co-I, DZYNE Technologies 2019–2021

Research Contributions (Graduate Researcher)

- **NSF CAREER: Learning and Using Models of Geo-Temporal Appearance** \$499,426
Co-I; PI: Nathan Jacobs (University of Kentucky) 2016–2021
- **NGA GeoLookbook: Modeling Worldwide Human Visual Appearance** \$299,204
Co-I; PI: Nathan Jacobs (University of Kentucky) 2014–2018
- **IARPA FINDER: Wide Area Localization of Depicted Objects (WALDO)** \$373,395
Co-I; PI: Nathan Jacobs (University of Kentucky) 2012–2016
- **DARPA ContextualEyes: A Context-Aware Surveillance System** \$743,131
Co-I; PI: Nathan Jacobs (University of Kentucky) 2011–2015

Mentorship

Selected individuals I have had the privilege of supporting as they worked toward their academic and professional goals.

Junior Researchers & Professional Staff

- Armin Hadzic DZYNE Technologies 2021–2025

Ph.D. Students

- Connor Greenwell University of Kentucky → Kitware 2022
- Hunter Blanton University of Kentucky → Yembo 2021
- M. Usman Rafique University of Kentucky → Zoox 2021
- Tawfiq Salem University of Kentucky → Purdue University 2019

M.S. Students

- Armin Hadzic University of Kentucky → DZYNE Technologies 2020
- Weilian Song University of Kentucky → Integrated Projects 2019
- Ryan Baltenberger University of Kentucky → Badger Technologies 2016

Undergraduate Research Students

• Weilian Song	University of Kentucky	2016–2019
• Connor Greenwell	University of Kentucky	2014–2016
• J. David Smith	University of Kentucky	2013–2015
• Ryan Baltenberger	University of Kentucky	2012–2015
• Jim Knochelmann	University of Kentucky	2011–2012

Selected Talks

- “Handling Image and Label Resolution Mismatch in Remote Sensing”, Jan. 2023, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa Village, HI.
- “Dynamic Traffic Modeling from Overhead Imagery”, Jun. 2020, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Virtual.
- “Computer Vision Applications of Deep Convolutional Neural Networks”, Nov. 2015, Keeping Current Seminar, University of Kentucky (Computer Science), Lexington, KY.
- “Activity Forecasting”, Nov. 2013, Birds of a Feather, University of Kentucky (Computer Science), Lexington, KY.
- “Camera Calibration using Atmospheric Cloud Motion”, Feb. 2013, Keeping Current Seminar, University of Kentucky (Computer Science), Lexington, KY.

Teaching

Teaching Assistant

- *Special Topics in Computer Science: Learning Based Methods for Computer Vision*, CS 685, (S2015), University of Kentucky
- *Introduction to Machine Learning*, CS 485, (F2013, F2014), University of Kentucky

Expertise and Skills

- **Machine Learning & AI:** Deep Learning, CNNs, Transformers/Attention, Generative AI (GANs, Diffusion Models), Self-/Weakly-Supervised Learning, Representation Learning, Transfer Learning
- **Computer Vision & 3D:** Geometric Vision, Camera Calibration, Multi-view Geometry, Pose/Depth Estimation, Image Geolocalization, Cross-View Learning/Synthesis, Scene Understanding, Semantic Segmentation
- **Sensor Data & Processing:** Satellite/Aerial/UAV Imagery, Panoramic/Street-Level, Webcam/Time-Lapse, Multispectral/Thermal, LiDAR, Audio, NLP/Text
- **Application Domains:** Geospatial/Geo-Temporal Analysis, Multimodal/Sensor Fusion, AR/VR, Image-Driven Mapping, Digital Forensics, Traffic Modeling, Remote Sensing, Change Detection
- **Languages & Tools:** Python, C/C++, CUDA, PyTorch (Lightning), TensorFlow/Keras, OpenCV, NumPy/SciPy, SQL, Bash/Shell, Dev (Vim, Jupyter), Agentic AI Tools (OpenCode, Antigravity, Ollama), Web (TypeScript, React), Geospatial (GDAL, Rasterio, PostGIS, GEE)
- **Systems & Infrastructure:** Linux/Unix, HPC (Slurm), Distributed Training (PyTorch DDP, Lightning), MLOps (TensorBoard, WandB, MLflow), Cloud (AWS), Containerization (Docker, Kubernetes), CI/CD, Git/DVC

- **Leadership & Strategy:** Technical Strategy & Roadmapping, Cross-Functional Leadership, R&D Management, Mentoring, Stakeholder Communication, Grant/Proposal Writing, Large-Scale Dataset Curation